# MOHAMMAD AKBAR BIN SHAH

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**Ph.D. Computer Science, Kansas State University · GPA 4.00/4.00 · Anticipated graduation: May 2030 · Available Summer 2027**

## EDUCATION

**Ph.D. in Computer Science — Kansas State University** *Aug 2025 – May 2030 (expected)*
GPA 4.00/4.00 | Manhattan, KS | Advisor: Dr. Doina Caragea, ML&DS Lab
Coursework: Programming Languages, Algorithms, Database Management Systems, Formal Language Theory

**B.Sc. in Computer Science & Engineering — Chittagong University of Engineering & Technology** *Feb 2019 – May 2024*
CGPA 3.71/4.00 | Chattogram, Bangladesh

## RESEARCH EXPERIENCE

**Graduate Research Assistant — ML&DS Lab, Kansas State University** *Jan 2026 – Present*
*NSF SaTC 2.0 project (Grant No. 2523438) | Manhattan, KS*

- **Infant distress recognition (ICTAI 2026, first author):** Built a **PyTorch** pipeline that fine-tunes **OpenFace-3.0** on infant faces and fuses it with **PyAFAR** action units or **MediaPipe** landmark geometry, reaching **0.91 weighted F1** on 19 unseen hospital patients vs. 0.73 unadapted; validated with subject-level bootstrap and ablations, and used **Grad-CAM** and **SHAP** to show the model attends to the facial regions clinicians use to score infant pain.
- **(Ongoing) Real-time tracheostomy-tube tracking:** Implementing Meta's **SAM 2/3** and **DINOv2** models for fully automatic, real-time tracking of an infant's tracheostomy tube to flag dislodgement.
- **(Ongoing) 3D bedside reconstruction:** Implementing Meta's **SAM 3D** models for periodic 3D reconstruction of the infant's body and surrounding objects, toward complete bedside awareness.

**Undergraduate Researcher — Chittagong University of Engineering & Technology** *May 2023 – May 2024*

- **Federated diabetic retinopathy grading (SN Computer Science, first author):** Implemented **Swin Transformer** federated learning with **FedAvg** and **FedProx** for five-stage classification across three heterogeneous datasets.
- Collected a private patient dataset from Bangladesh Eye Hospital to test on an unseen site; reached **76.8%** accuracy (3.7% above prior work) and **78.0%** on the unseen hospital's patients.

## PUBLICATIONS

- **M. A. B. Shah**, A. Zareian, V. Bondalakunta, N. Haque, S. Wei, X. Wan, M. Dunham, G. Amariucai, D. Caragea. "Multi-Representation Facial Feature Fusion for Cross-Domain Infant Distress Recognition." *IEEE ICTAI*, 2026 (accepted).
  *Infant-specific adaptation drives cross-hospital transfer; fusing a second facial representation adds a further gain.*
- V. Bondalakunta, A. Zareian Jahromi, **M. A. B. Shah**, N. Haque, G. T. Amariucai, D. Caragea, S. Wei, X. Wan, M. E. Dunham. "A Privacy-Preserving Framework for Sharing and Learning from Scarce Medical Video." *HealthSec Workshop*, 2026 (accepted).
  *Face-swap anonymization so hospitals can share patient video; off-the-shelf swappers still leak identity.*
- A. Zareian Jahromi, V. Bondalakunta, **M. A. B. Shah**, N. Haque, S. Wei, G. T. Amariucai. "Picture the Epsilon: Pursuing Identity-Level Privacy Guarantees for Images." *arXiv:2608.17147*, 2026.
  *Compares four identity-level privacy audits for face generators; only a held-out classifier test is valid.*
- **M. A. B. Shah**, T. Azhar, Asaduzzaman, M. Chowdhury. "A Swin Transformer-Based Federated Learning Approach for Classification of Diabetic Retinopathy." *SN Computer Science*, 2025.
  *Grades diabetic retinopathy across hospitals without sharing patient images; tested on an unseen hospital.*

## SELECTED PROJECT

**PuffCut — Automated Extrudate Measurement** *Fall 2026*
*Python, PyTorch, SAM 2.1, OpenCV, NumPy, SciPy, pandas*

- Built a pipeline that finds every knife cut in 240 fps extruder video and automatically prompts **SAM 2.1** to segment and track each newly cut corn puff, with no manual labeling.
- Measures each piece's size and expansion in millimetres; 244 cuts found in a 10.7 s clip. Code

## CERTIFICATIONS

- **Human Subjects Research (IRB)**, CITI Program, 2025 — IRB review, informed consent and protection of research participants.
- **Responsible Conduct of Research**, CITI Program, 2025 — data management, reproducibility, authorship and research integrity.
- **Export Compliance**, CITI Program, 2025 — U.S. export controls on research data, software and international collaboration.

## TEACHING & INDUSTRY EXPERIENCE

**Graduate Teaching Assistant — Kansas State University** (CIS 115 Introduction to Computing Science) *Aug 2025 – Dec 2025*
**Lecturer — East Delta University** (Data Structures, Computer Networks, Microprocessor & Interfacing) *Jan 2025 – Jul 2025*
**Industrial Attachment — Upay, UCB Fintech** (credit-scoring model with logistic regression; React and Django app) *Sep 2023 – Oct 2023*

## TECHNICAL SKILLS

**Programming Languages:** Python, C, C++, SQL (MySQL, PostgreSQL)
**Systems & Tools:** Linux, Docker, Git, Conda; OpenCV, OpenFace-3.0, MediaPipe, PyAFAR; React, Django
**ML & Algorithms:** PyTorch, Hugging Face Transformers, NumPy, scikit-learn, pandas; deep learning, computer vision, transfer learning, self-supervised and federated learning; data structures and algorithms; interpretability (SHAP, Grad-CAM)